

Hardware Connection Guide

AVR (ATmega328P) ↔ PCF85176 ↔ S5026 Clover LCD

STATIC DRIVE MODE — 32 direct segment lines (S0–S31)

4 digits × 7 segs = 28 + 3 decimal points + 1 colon = 32 segments

Based on NXP PCF85176 Datasheet Rev. 5.1 — September 2021

1. Introduction & Design Choice

This document specifies the connections to drive the **Clover Display S5026** 4-digit 7-segment LCD using the **NXP PCF85176** in **STATIC drive mode**. In static mode, the PCF85176 uses a single backplane (BP0) and dedicates each of its 40 segment outputs (S0–S39) to drive one individual LCD segment directly. This gives full independent control over every segment simultaneously — no time-division multiplexing.

The S5026 display has the following segments requiring individual drive lines:

Display Element	Count	PCF85176 Mapping
4 digits × 7 segments (a–g)	28 lines	S0–S27
3 decimal points (one per digit, except last may vary)	3 lines	S28–S30
1 colon symbol	1 line	S31
Total required	32 lines	S0–S31
Remaining unused	8 lines	S32–S39 — left unconnected

Why static mode is the right choice here: With 32 independent segment lines, every segment is always driven at full VLCD amplitude — giving maximum contrast, no ghosting, and zero firmware overhead for multiplexing. The PCF85176 was specifically designed to support static drive with up to 40 segments on one backplane (Table 12, NXP datasheet).

2. Static Drive Mode — How It Works

In static drive mode the PCF85176 uses **only BP0** as the single backplane. BP0 carries a continuous AC square wave at the frame frequency. Each Sn output is individually switched to be either in-phase with BP0 (segment OFF) or anti-phase with BP0 (segment ON). This produces a net AC voltage across the LCD glass only for the ON segments.

The discrimination ratio in static mode is **infinite** — a segment that is off has exactly 0V RMS across it, and a segment that is on has exactly VLCD across it. This is the best possible contrast achievable with an LCD.

2.1 mode-set Command Byte for Static Mode

Bit Field	Value	Meaning
Bit 7 (C)	0	Continuation = 0 → last command byte; next bytes are data
Bits 6–5	10	Fixed value — always 10 binary per NXP datasheet
Bit 4	0	Unused – set to 0
Bit 3 (E)	1	Enable display (0 = blank)
Bit 2 (B)	x	Bias — not applicable in static mode (value ignored)
Bits 1–0 (M[1:0])	01	Drive mode = STATIC; only BP0 active

Mode-set byte = 0x49 (binary: 0 1 0 0 1 0 0 1)
 C=0, fixed=10, unused=0, E=1 (enabled), B=0, M[1:0]=01 (static)
 Send this byte first over I²C after the >1 ms power-on delay.

3. Complete Connection Table

3.1 Power Supply

Connection	Connects To	Notes
VDD (TSSOP pin 48)	AVR +5V	Logic supply 1.8–5.5V; match AVR voltage
VSS (TSSOP pin 54)	GND	Common ground — connect to AVR GND
VLCD (TSSOP pin 55)	+5V (= VDD)	LCD drive voltage. Tie to VDD for standard TN LCD
100 nF capacitor	VDD pin to GND	Decoupling — place physically close to pin 48
100 nF capacitor	VLCD pin to GND	Decoupling — place physically close to pin 55
10 µF capacitor	VLCD rail to GND	Bulk decoupling on the +5V VLCD rail

3.2 I²C Bus — AVR ↔ PCF85176

Connection	Connects To	Notes
SDA (TSSOP pin 44)	AVR PC4 (Arduino A4/SDA)	4.7 kΩ pull-up to +5V required
SCL (TSSOP pin 45)	AVR PC5 (Arduino A5/SCL)	4.7 kΩ pull-up to +5V required
4.7 kΩ resistor	SDA line → VDD (+5V)	I ² C bus pull-up (open-drain bus)
4.7 kΩ resistor	SCL line → VDD (+5V)	I ² C bus pull-up (open-drain bus)

I²C address (7-bit): Fixed bits 0 1 1 1 0, then SA0.
 SA0 = GND → address = **0x38** (recommended default)
 SA0 = VDD → address = **0x3A**
 PCF85176 is write-only. No read transactions needed.

3.3 Address & Configuration Pins

Pin	Connect To	Reason
SA0 (TSSOP pin 53)	GND	Sets I ² C address to 0x38
A0 (TSSOP pin 50)	GND	Subaddress bit 0 — tie low for single device
A1 (TSSOP pin 51)	GND	Subaddress bit 1 — tie low for single device
A2 (TSSOP pin 52)	GND	Subaddress bit 2 — tie low for single device
OSC (TSSOP pin 49)	GND (VSS)	MUST be tied to GND to enable internal oscillator
CLK (TSSOP pin 47)	Leave open (float)	External clock — not used when OSC=GND
SYNC(TSSOP pin 46)	Leave open (float)	Cascade sync — not used for single IC

3.4 Backplane Connection — STATIC MODE

Critical difference from multiplexed mode: In static drive, only **BP0** is connected to the LCD backplane electrode. BP1, BP2, BP3 are left completely open-circuit (not connected to anything). The LCD panel itself has a single backplane rail (common electrode) which connects to BP0.

PCF85176 Pin	Connects To	Notes
BP0 (TSSOP pin 56)	S5026 LCD backplane (COM)	ONLY backplane used in static mode
BP1 (TSSOP pin 1)	Leave open	Not connected — static mode uses BP0 only
BP2 (TSSOP pin 2)	Leave open	Not connected — static mode uses BP0 only
BP3 (TSSOP pin 3)	Leave open	Not connected — static mode uses BP0 only

4. Segment-to-Pin Mapping (S0–S31)

Each of the 32 segment wires connects one PCF85176 output pin directly to one segment electrode on the S5026 LCD glass. The table below gives the logical mapping. The physical pin number on S5026 must be confirmed from the Clover COG datasheet for your specific part variant, but the segment names (a–g, dp, colon) are standard.

PCF85176 Segment	TSSOP56 Phys. Pin	LCD Element (S5026)
S0	Pin 4	Digit 1 — segment a
S1	Pin 5	Digit 1 — segment b
S2	Pin 6	Digit 1 — segment c
S3	Pin 7	Digit 1 — segment d
S4	Pin 8	Digit 1 — segment e
S5	Pin 9	Digit 1 — segment f
S6	Pin 10	Digit 1 — segment g

PCF85176 Segment	TSSOP56 Phys. Pin	LCD Element (S5026)
S7	Pin 11	Digit 2 — segment a
S8	Pin 12	Digit 2 — segment b
S9	Pin 13	Digit 2 — segment c
S10	Pin 14	Digit 2 — segment d
S11	Pin 15	Digit 2 — segment e
S12	Pin 16	Digit 2 — segment f
S13	Pin 17	Digit 2 — segment g
S14	Pin 18	Digit 3 — segment a
S15	Pin 19	Digit 3 — segment b
S16	Pin 20	Digit 3 — segment c
S17	Pin 21	Digit 3 — segment d
S18	Pin 22	Digit 3 — segment e
S19	Pin 23	Digit 3 — segment f
S20	Pin 24	Digit 3 — segment g
S21	Pin 25	Digit 4 — segment a
S22	Pin 26	Digit 4 — segment b
S23	Pin 27	Digit 4 — segment c
S24	Pin 28	Digit 4 — segment d
S25	Pin 29	Digit 4 — segment e
S26	Pin 30	Digit 4 — segment f
S27	Pin 31	Digit 4 — segment g
S28	Pin 32	Digit 1 — decimal point (dp1)
S29	Pin 33	Digit 2 — decimal point (dp2)
S30	Pin 34	Digit 3 — decimal point (dp3)
S31	Pin 35	Colon symbol

Segment ordering note: The mapping above assigns S0–S6 to digit 1 (a–g), S7–S13 to digit 2 (a–g), S14–S20 to digit 3 (a–g), S21–S27 to digit 4 (a–g), S28–S30 to the three decimal points, and S31 to the colon. This is a logical assignment — you can reorder these to match your S5026 PCB trace routing. What matters is that you update your firmware lookup table to reflect whatever physical mapping you choose.

5. Full TSSOP56 Pin Disposition Summary

Quick reference of every pin on the PCF85176T (TSSOP56) and what it connects to in this static-drive design.

TSSOP56 Pin	Signal	Connection
1	BP2	OPEN — unused in static mode
2	BP1	OPEN — unused in static mode
3	BP3	OPEN — unused in static mode
4	S0	LCD segment a — Digit 1
5	S1	LCD segment b — Digit 1
6	S2	LCD segment c — Digit 1
7	S3	LCD segment d — Digit 1
8	S4	LCD segment e — Digit 1
9	S5	LCD segment f — Digit 1
10	S6	LCD segment g — Digit 1
11	S7	LCD segment a — Digit 2
12	S8	LCD segment b — Digit 2
13	S9	LCD segment c — Digit 2
14	S10	LCD segment d — Digit 2
15	S11	LCD segment e — Digit 2
16	S12	LCD segment f — Digit 2
17	S13	LCD segment g — Digit 2
18	S14	LCD segment a — Digit 3
19	S15	LCD segment b — Digit 3
20	S16	LCD segment c — Digit 3
21	S17	LCD segment d — Digit 3
22	S18	LCD segment e — Digit 3
23	S19	LCD segment f — Digit 3
24	S20	LCD segment g — Digit 3
25	S21	LCD segment a — Digit 4
26	S22	LCD segment b — Digit 4
27	S23	LCD segment c — Digit 4
28	S24	LCD segment d — Digit 4

TSSOP56 Pin	Signal	Connection
29	S25	LCD segment e — Digit 4
30	S26	LCD segment f — Digit 4
31	S27	LCD segment g — Digit 4
32	S28	LCD decimal point — Digit 1 (dp1)
33	S29	LCD decimal point — Digit 2 (dp2)
34	S30	LCD decimal point — Digit 3 (dp3)
35	S31	LCD colon symbol
36	S32	OPEN — unused
37	S33	OPEN — unused
38	S34	OPEN — unused
39	S35	OPEN — unused
40	S36	OPEN — unused
41	S37	OPEN — unused
42	S38	OPEN — unused
43	S39	OPEN — unused
44	SDA	AVR PC4 (A4/SDA) + 4.7 kΩ to +5V
45	SCL	AVR PC5 (A5/SCL) + 4.7 kΩ to +5V
46	SYNC	OPEN — not used
47	CLK	OPEN — using internal oscillator
48	VDD	+5V (+ 100 nF to GND)
49	OSC	GND — enables internal oscillator
50	A0	GND
51	A1	GND
52	A2	GND
53	SA0	GND → I ² C address 0x38
54	VSS	GND
55	VLCD	+5V (+ 100 nF to GND, + 10 μF to GND)
56	BP0	S5026 LCD backplane (COM electrode)

6. I²C Protocol & AVR Code Skeleton

In static mode each RAM address (0–39) maps directly to one segment output (S0–S39). Writing a nibble value to RAM address N controls Sn: bit 0 of the nibble = segment state (1 = ON, 0 = OFF for BP0 in static mode). Send 32 data bytes to update all 32 active segments in one I²C transaction.

```
#include

#define PCF_ADDR 0x38 // SA0=GND
#define MODE_STATIC 0x49 // C=0, 10, E=1, B=0, M=01 (static)

// segment bitmasks: index = segment S0..S31
// 1 = ON (segment lit), 0 = OFF
uint8_t segs[40]; // 40 nibbles, only [0..31] used

void pcf_init() {
  Wire.begin();
  delay(2); // >1 ms POR delay (NXP requirement)
  Wire.beginTransmission(PCF_ADDR);
  Wire.write(MODE_STATIC); // mode-set: static drive, display ON
  Wire.endTransmission();
}

void pcf_update() {
  // Load-data-pointer to address 0, C=1 so next byte is data
  Wire.beginTransmission(PCF_ADDR);
  Wire.write(0x80); // load-data-pointer: C=1, P=0
  for (uint8_t i = 0; i < 32; i++) {
    Wire.write(segs[i] & 0x01); // 1 nibble per segment
  }
  Wire.endTransmission();
}

// Example: display '1' on Digit 1 (segments b and c ON)
void display_one_digit1() {
  memset(segs, 0, sizeof(segs));
  segs[1] = 1; // S1 = seg b, Digit 1
  segs[2] = 1; // S2 = seg c, Digit 1
  pcf_update();
}
```

7. Pre-Power-On Checklist

- VDD (pin 48) connected to +5V with 100 nF decoupling cap to GND

- VSS (pin 54) connected to GND
- VLCD (pin 55) connected to +5V with 100 nF + 10 μ F decoupling to GND
- OSC (pin 49) tied to GND — enables internal oscillator
- SA0 (pin 53), A0 (pin 50), A1 (pin 51), A2 (pin 52) — all to GND
- CLK (pin 47) — left open (floating)
- SYNC (pin 46) — left open (floating)
- SDA (pin 44) → AVR PC4 with 4.7 k Ω pull-up to +5V
- SCL (pin 45) → AVR PC5 with 4.7 k Ω pull-up to +5V
- BP0 (pin 56) → S5026 LCD single backplane (COM) electrode
- BP1 (pin 1), BP2 (pin 2), BP3 (pin 3) — all left open
- S0–S6 (pins 4–10) → S5026 segments a–g of Digit 1
- S7–S13 (pins 11–17) → S5026 segments a–g of Digit 2
- S14–S20 (pins 18–24) → S5026 segments a–g of Digit 3
- S21–S27 (pins 25–31) → S5026 segments a–g of Digit 4
- S28 (pin 32) → S5026 decimal point Digit 1
- S29 (pin 33) → S5026 decimal point Digit 2
- S30 (pin 34) → S5026 decimal point Digit 3
- S31 (pin 35) → S5026 colon symbol
- S32–S39 (pins 36–43) — all left open (unused)
- No resistors between PCF85176 and LCD glass (AC drive, no DC current)
- Firmware sends mode-set 0x49 after ≥ 2 ms power-on delay
- S5026 physical pin numbers for each segment verified from Clover COG spec

Reference: NXP PCF85176 Product Data Sheet Rev. 5.1, 15 September 2021. Static mode: Table 12 (1 backplane = 40 segments max), Table 13 (discrimination = ∞), Table 6 (mode-set M[1:0]=01 for static). Physical segment-to-S-pin assignment on S5026 must be confirmed from Clover Display COG specification for the S5026 part number.